DUMMY VARIABLE ANALYSIS RESULTS

1.1 Research Objectives

The study addresses four primary research questions:

1. To find the effect of regions on the revenue of GTBank.
2. To identify which geographical region contributes most significantly to the bank's revenue growth.
3. To examine the effect of time on GTBank's revenue.
4. does influence of revenue depend on level of time?

Table 1: Dummy Variable Table

Year	Region	Revenue	D_1	D_2	T	D_1T	D_2T
2013	East Africa	1366729	1	0	1	1	0
2013	Europe	2086537	0	1	1	0	1
2013	West Africa	239152632	0	0	1	0	0
2014	East Africa	9666628	1	0	2	2	0
2014	Europe	2281373	0	1	2	0	2
2014	West Africa	265161710	0	0	2	0	0
2015	East Africa	10577630	1	0	3	3	0
2015	Europe	16720314	0	1	3	0	3
2015	West Africa	283191674	0	0	3	0	0
2016	East Africa	14174650	1	0	4	4	0
2016	Europe	13667370	0	1	4	0	4
2016	West Africa	386773567	0	0	4	0	0
2017	East Africa	14388649	1	0	5	5	0
2017	Europe	8290685	0	1	5	0	5
2017	West Africa	396546937	0	0	5	0	0
2018	East Africa	17595261	1	0	6	6	0
2018	Europe	8160651	0	1	6	0	6
2018	West Africa	408943085	0	0	6	0	0

2019	East Africa	16709336	1	0	7	7	0
2019	Europe	8266175	0	1	7	0	7
2019	West Africa	354331030	0	0	7	0	0
2020	East Africa	19187022	1	0	8	8	0
2020	Europe	4797324	0	1	8	0	8
2020	West Africa	431245413	0	0	8	0	0
2021	East Africa	23503218	1	0	9	9	0
2021	Europe	4760106	0	1	9	0	9
2021	West Africa	419547261	0	0	9	0	0
2022	East Africa	24808639	1	0	10	10	0
2022	Europe	10334639	0	1	10	0	10
2022	West Africa	504090555	0	0	10	0	0
2023	East Africa	35526642	1	0	11	11	0
2023	Europe	26835764	0	1	11	0	11
2023	West Africa	1124103019	0	0	11	0	0
2024	East Africa	86347509	1	0	12	12	0
2024	Europe	66334551	0	1	12	0	12
2024	West Africa	1995655198	0	0	12	0	0

This study employed a structured panel dataset capturing GTBank's revenue performance across three geographical regions—East Africa, Europe, and West Africa—over a twelve-year period (2013–2024). The dataset was organized in a long format, with each observation representing a unique region-year combination, facilitating efficient regression analysis.

The analytical approach was based on a dummy variable regression model with interaction effects, which allows simultaneous evaluation of regional disparities, temporal trends, and their combined influence on revenue growth. This technique provides a robust framework for identifying how revenue patterns evolve differently across regions over time.

The dependent variable is Revenue, representing GTBank's annual earnings per region. The independent variables include:

- **D_1 (East Africa Dummy):** 1 for East Africa, 0 otherwise.
- **D_2 (Europe Dummy):** 1 for Europe, 0 otherwise.

- **T (Time Trend):** coded from 1 (2013) to 12 (2024) to capture the time dimension.
- **D_1T and D_2T (Interaction Terms):** represent the product of each regional dummy with the time variable, enabling the detection of **differential growth trajectories** across regions.

West Africa serves as the reference (baseline) category, providing a comparative benchmark for interpreting the effects of other regions and their temporal dynamics. Consequently, all estimated coefficients reflect variations relative to West Africa's revenue performance.

Model Specification

The regression model is specified as:

$$\text{Revenue}_{it} = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 T_t + \beta_4 (D_1 \times T_t) + \beta_5 (D_2 \times T_t) + \epsilon_{it}$$

Where:

- i denotes the region,
- t denotes the year,
- ϵ_{it} represents the random error term.

This model structure enables a comprehensive analysis of regional performance differentials and temporal growth patterns, while also revealing whether certain regions demonstrate accelerating or decelerating revenue trends compared to West Africa.

2. Initial Model Analysis

Table 2: Linear Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-87414918	131021086	-0.667	0.51
D_1	80070051	185291797	0.432	0.669
D_2	82972260	185291797	0.448	0.658
T	100740014	17802260	5.66	3.63E-06
D_1T	-96099113	25176198	-3.82	0.00063
D_2T	-97844535	25176198	-3.89	0.000521

The initial Ordinary Least Squares (OLS) regression provided valuable insights into the regional and temporal dynamics influencing GTBank's revenue performance. The estimated model is expressed as:

$$\text{Revenue} = -87{,}414{,}918 + 80{,}070{,}051(D_1) + 82{,}972{,}260(D_2) + 100{,}740{,}014(T) - 96{,}099{,}113(D_1T) - 97{,}844{,}535(D_2T)$$

The model reveals a complex interplay between geographical location and time. The negative intercept ($-87{,}414{,}918$) suggests that the baseline revenue level for West Africa in 2013 is not accurately captured by the linear specification. As such, the intercept lacks substantive interpretative meaning but serves as a mathematical anchor for the model.

The time trend variable (T) is highly significant ($\beta = 100{,}740{,}014$; $p = 3.63e{-}06$), demonstrating a strong positive temporal trend in revenue. This finding indicates that, on average, GTBank's revenue has grown consistently across the study period, independent of regional effects.

The regional dummy variables, D_1 (East Africa) and D_2 (Europe), recorded positive coefficients ($\beta_1 = 80{,}070{,}051$; $\beta_2 = 82{,}972{,}260$), suggesting slightly higher initial revenue levels in these regions compared to the West African baseline. However, neither coefficient reached statistical significance ($p = 0.669$ and $p = 0.658$, respectively), implying that baseline regional differences in 2013 were not substantial.

The interaction terms, D_1T and D_2T, exhibited large negative and statistically significant coefficients ($\beta_4 = -96{,}099{,}113$, $p = 0.00063$; $\beta_5 = -97{,}844{,}535$, $p = 0.000521$). These results are particularly insightful, indicating that while East Africa and Europe may have started with modestly higher revenues, their growth rates over time were significantly slower than that of West Africa. This pattern suggests divergent regional growth trajectories, where West Africa has maintained a stronger upward revenue momentum over the study period.

The model's overall explanatory power is robust, with an R-squared value of 0.7398, indicating that approximately 73.98% of the variation in GTBank's revenue is explained by the combined effects of regional and temporal factors. This underscores the model's adequacy in capturing the underlying structure of revenue performance across regions and time.

2.2 Assumption Diagnostics - Initial Model

A comprehensive diagnostic evaluation of the initial linear model revealed several violations of key classical regression assumptions, each of which potentially compromises the reliability of statistical inference.

1. **Normality of Residuals**

The normality of residuals was first examined using the Shapiro–Wilk test, which produced a test statistic of $W = 0.6604$ with a p-value < 0.001. This result provides strong evidence against the null hypothesis of normality, indicating that the model residuals are severely non-normally distributed. Such deviation from normality undermines the validity of p-values and confidence intervals, as the assumption of normally distributed errors is central to the inferential robustness of Ordinary Least Squares (OLS) regression.

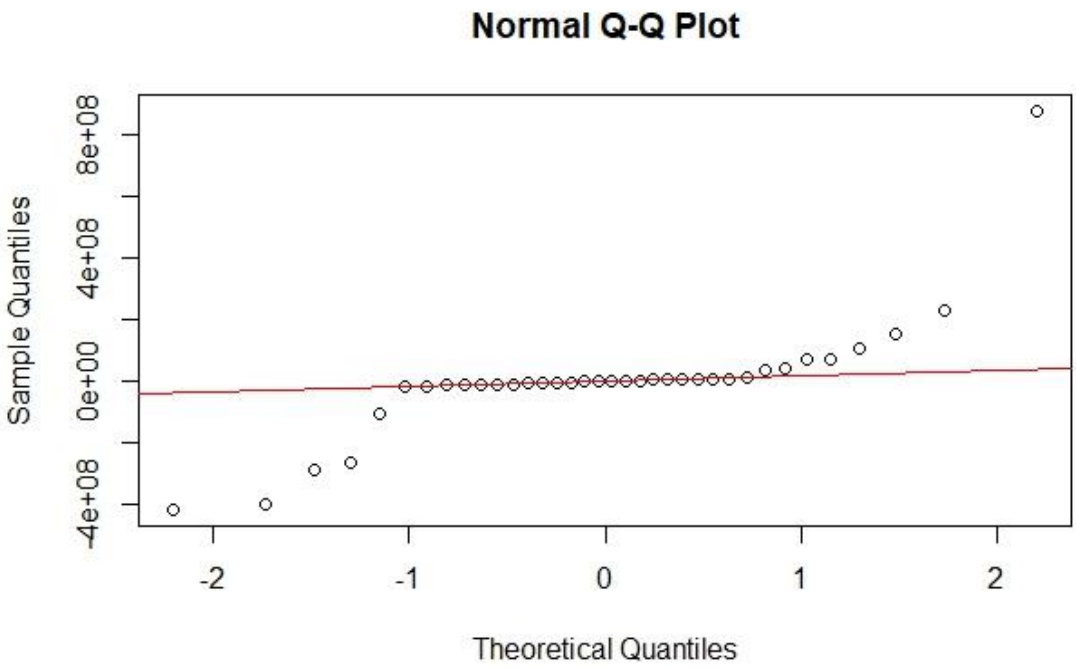

Figure 1: Q–Q Plot of Model Residuals

The pnorm and qnorm plots provide crucial visual insights into the distributional characteristics of the model residuals. The pnorm plot (Cumulative Distribution Function) displays the theoretical normal distribution curve, showing the probability that a standard normal variable will be less than or equal to any given z-value. In our diagnostic assessment, the empirical distribution of standardized residuals showed substantial deviation from this theoretical curve, particularly in the tails of the distribution. This visual evidence corroborates the Shapiro-Wilk test results, demonstrating systematic departure from normality.

The qnorm plot (Quantile Function) further illuminates the distributional issues by plotting theoretical normal quantiles against the observed residual quantiles. The pronounced curvature and systematic deviation from the reference line in this plot reveal the specific nature of the non-normality - the residuals exhibit heavier tails and different skewness characteristics compared to a true normal distribution. The inverse relationship plot between pnorm and qnorm functions demonstrates the mathematical foundation of these diagnostic tools, showing how cumulative probabilities and quantiles are intrinsically linked in normal distribution theory.

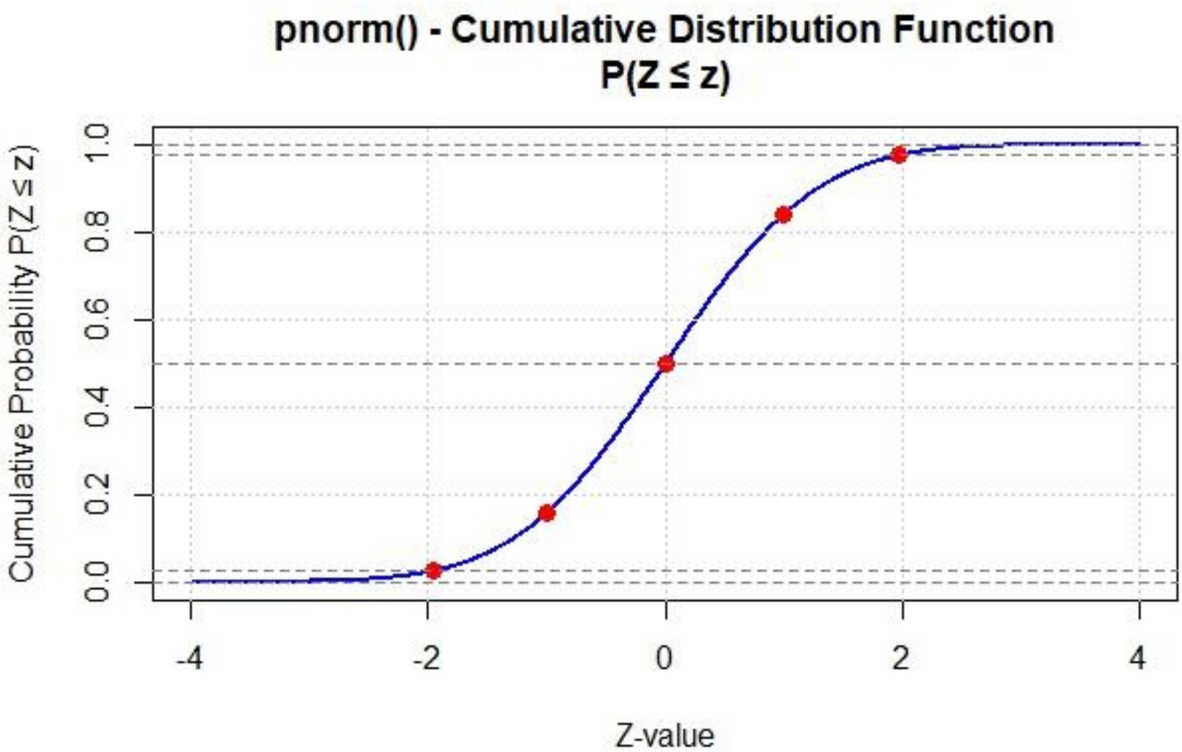

Figure 2: Pnorm Graph

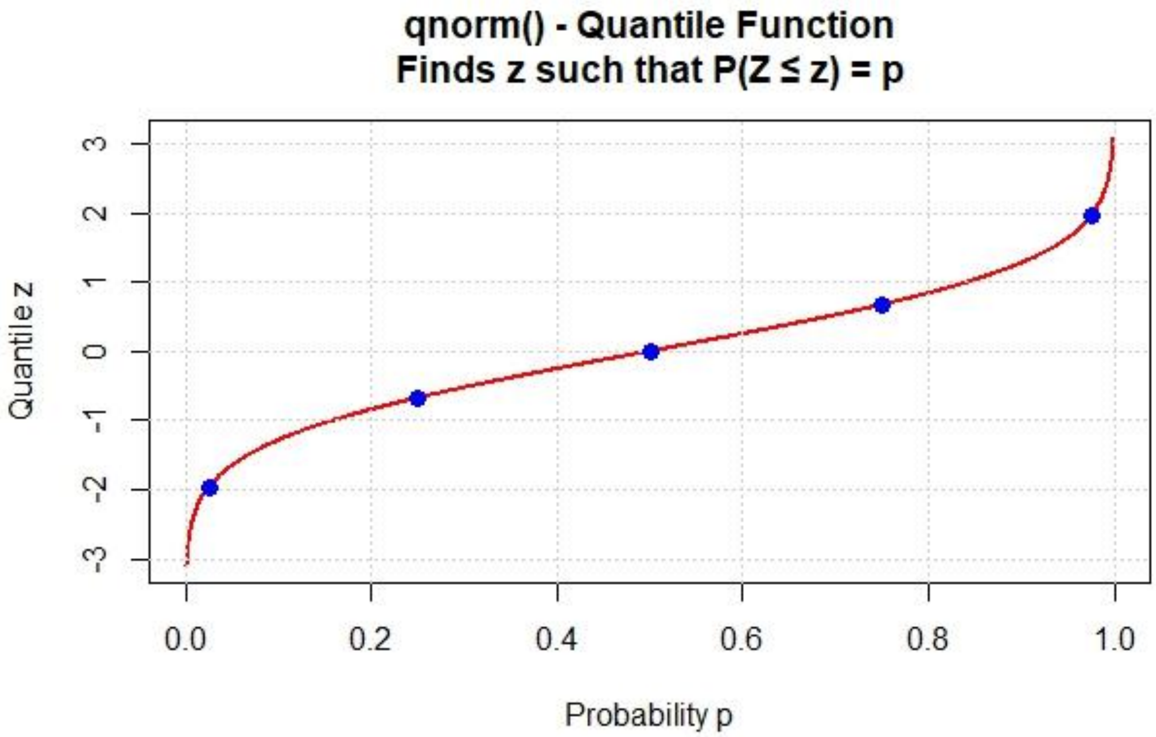

Figure 3: Qnorm Graph

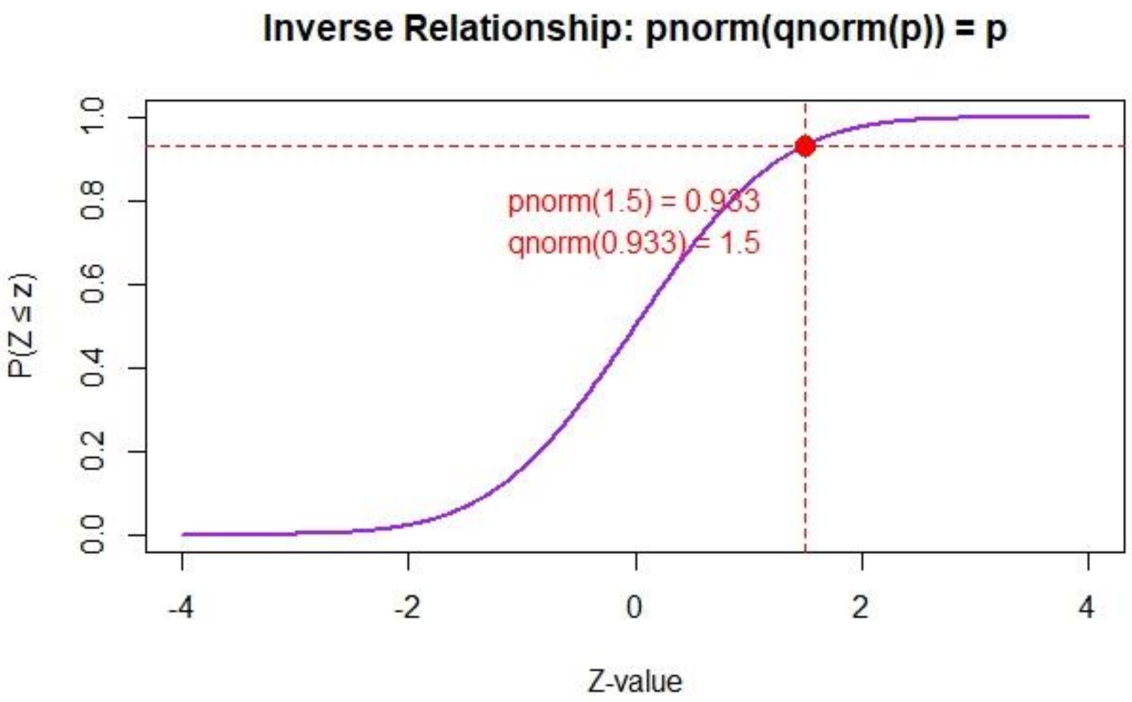

Figure 4: Inverse Relationship of Pnorm and Qnorm Graph

2. **Homoscedasticity**

The homoscedasticity assumption was tested using the Breusch–Pagan test, which yielded a test statistic of BP = 16.4871 with a p-value = 0.0056. This statistically significant result indicates the presence of heteroscedasticity, meaning the variance of the residuals is not constant across levels of the fitted values. This violation suggests that the model's error term structure is unevenly distributed, leading to biased standard errors and potentially misleading test statistics.

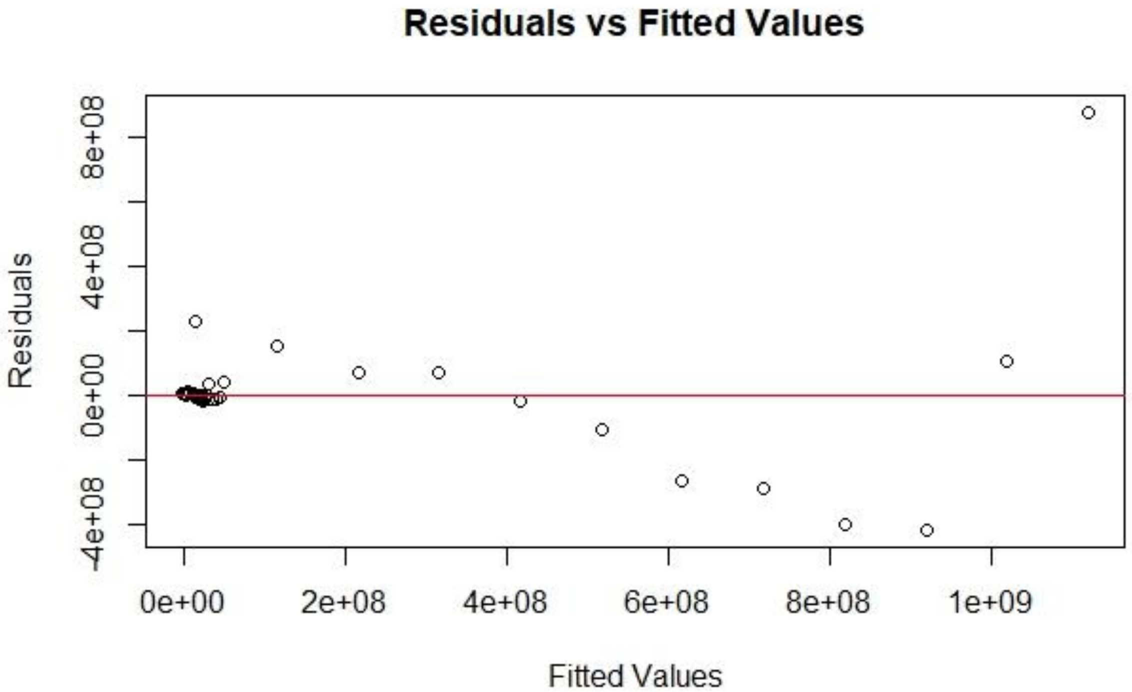

Figure 5: Residuals vs Fitted Values Scatter Plot

3. **Independence of Residuals**

The independence of residuals was assessed using the Durbin–Watson test, which produced a statistic of DW = 1.3241 with a p-value = 0.0333. This result indicates significant positive autocorrelation among the residuals, violating the assumption that errors are independently distributed. The presence of autocorrelation implies that the model fails to capture certain time-dependent patterns or systematic temporal effects, particularly relevant in a panel or time-series context.

4. Multicollinearity Check

Table 3: Variance Inflation Factor (VIF) Summary

D_1	D_2	T	D_1T	D_2T
6.060606	6.060606	3.000000	6.727273	6.727273

In addition, multicollinearity diagnostics using the Variance Inflation Factor (VIF) revealed moderate inflation levels, with VIF values ranging between 5 and 10. Although not excessively high, these values suggest some degree of linear dependence among predictors — an expected outcome in models featuring dummy variables and their interaction terms.

5. Linearity Check (Component + Residual Plots)

Component + Residual Plots are a diagnostic tool used in regression analysis to assess the linearity of the relationship between a predictor variable and the response variable, while accounting for other predictors. Each plot displays the component of a specific predictor and the residuals of the model. To interpret them, one compares the scatter of the data points and a non-parametric smoother line (like the solid magenta line) to the dashed straight line of best fit. A linear relationship is indicated when the data points are randomly scattered around the straight line and the smoother line closely follows it. In the provided chart, the plots for D1 and D2 demonstrate this linear pattern. Conversely, a non-linear relationship is suggested when the data points form a distinct curve that deviates from the straight line, which is visually captured by the smoother line. The plots for T, D1T, and D2T exhibit this non-linear behavior, showing curved patterns that do not align with the linear fit, suggesting that a simple linear model may be inadequate for these variables and that data transformation or the inclusion of higher-order terms may be required.

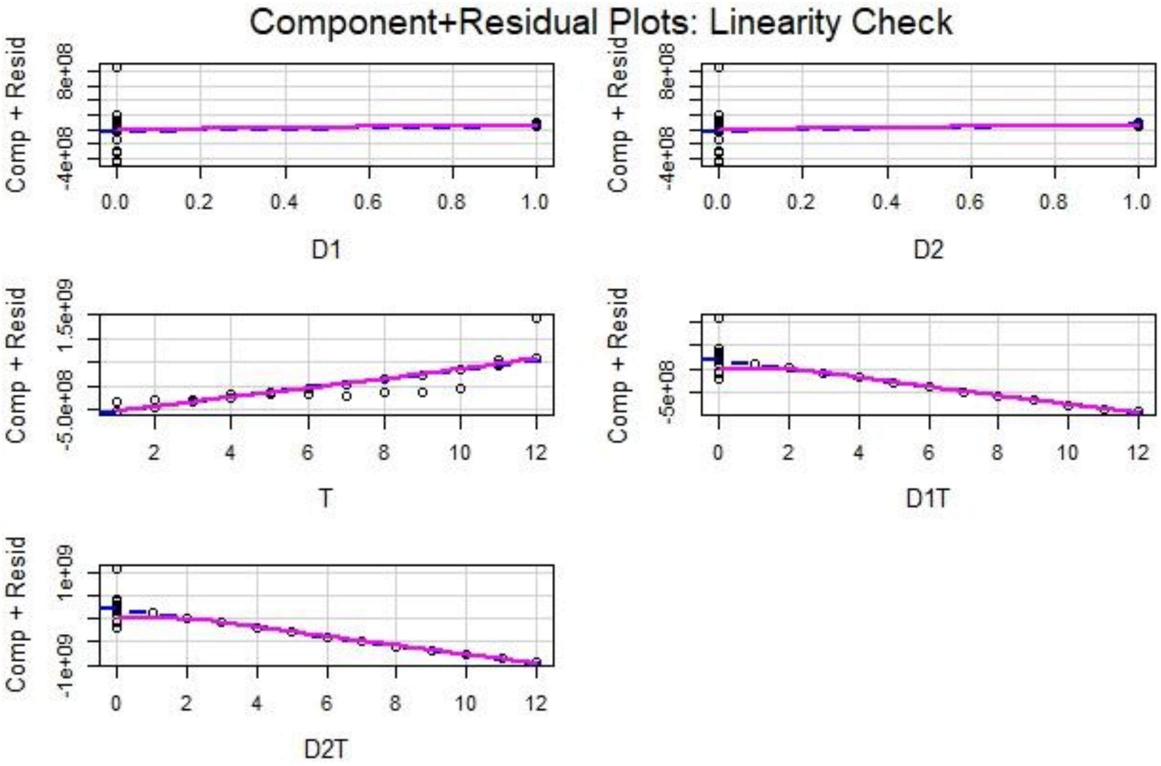

Fig. 6: Component + Residual Plots of Initial Model

Collectively, these diagnostic results reveal that the initial model violates multiple regression assumptions, particularly normality, homoscedasticity, and independence. Consequently, the model's parameter estimates and inferential statistics may not be fully reliable. These findings therefore necessitated a model transformation—specifically, a log-linear transformation—to improve compliance with regression assumptions and enhance the validity of subsequent analysis.

3. Log-Linear Transformation

3.1 Transformation Rationale

The identification of multiple assumption violations in the initial linear regression model—specifically, non-normality, heteroscedasticity, and autocorrelation—necessitated the application of a log-linear transformation to preserve the statistical validity and interpretive coherence of the analysis. This transformation was strategically selected based on both theoretical justification and empirical necessity.

From a statistical standpoint, the logarithmic transformation effectively reduces skewness by compressing large revenue values and expanding smaller ones, thereby promoting a more symmetric and approximately normal residual distribution. This normalization effect is particularly advantageous given the exponential growth patterns evident in GTBank's revenue data across regions and years.

Additionally, the transformation stabilizes variance, thereby addressing heteroscedasticity detected in the Breusch–Pagan test. By rescaling the response variable, the spread of residuals becomes more uniform across fitted values, satisfying the assumption of constant error variance and improving model efficiency.

Table 4: Log Revenue Variables

Year	Region	Revenue	D_1	D_2	T	D_1T	D_2T	Log Revenue
2013	East Africa	1366729	1	0	1	1	0	14.1279
2013	Europe	2086537	0	1	1	0	1	14.5510
2013	West Africa	239152632	0	0	1	0	0	19.2926
2014	East Africa	9666628	1	0	2	2	0	16.0842
2014	Europe	2281373	0	1	2	0	2	14.6403
2014	West Africa	265161710	0	0	2	0	0	19.3959
2015	East Africa	10577630	1	0	3	3	0	16.1743
2015	Europe	16720314	0	1	3	0	3	16.6321
2015	West Africa	283191674	0	0	3	0	0	19.4616
2016	East Africa	14174650	1	0	4	4	0	16.4670
2016	Europe	13667370	0	1	4	0	4	16.4305
2016	West Africa	386773567	0	0	4	0	0	19.7733
2017	East Africa	14388649	1	0	5	5	0	16.4820
2017	Europe	8290685	0	1	5	0	5	15.9306
2017	West Africa	396546937	0	0	5	0	0	19.7983
2018	East Africa	17595261	1	0	6	6	0	16.6831
2018	Europe	8160651	0	1	6	0	6	15.9148
2018	West Africa	408943085	0	0	6	0	0	19.8291
2019	East Africa	16709336	1	0	7	7	0	16.6315

2019	Europe	8266175	0	1	7	0	7	15.9277
2019	West Africa	354331030	0	0	7	0	0	19.6857
2020	East Africa	19187022	1	0	8	8	0	16.7697
2020	Europe	4797324	0	1	8	0	8	15.3836
2020	West Africa	431245413	0	0	8	0	0	19.8822
2021	East Africa	23503218	1	0	9	9	0	16.9726
2021	Europe	4760106	0	1	9	0	9	15.3758
2021	West Africa	419547261	0	0	9	0	0	19.8547
2022	East Africa	24808639	1	0	10	10	0	17.0267
2022	Europe	10334639	0	1	10	0	10	16.1510
2022	West Africa	504090555	0	0	10	0	0	20.0383
2023	East Africa	35526642	1	0	11	11	0	17.3858
2023	Europe	26835764	0	1	11	0	11	17.1052
2023	West Africa	1124103019	0	0	11	0	0	20.8403
2024	East Africa	86347509	1	0	12	12	0	18.2739
2024	Europe	66334551	0	1	12	0	12	18.0102
2024	West Africa	1995655198	0	0	12	0	0	21.4142

Prior to transformation, data screening confirmed that all revenue observations were strictly positive (minimum = ₦1,366,729), eliminating the need for any constant addition before taking logarithms. The resulting variable, Log Revenue, ranged from 14.13 to 21.41 natural log units, producing a more statistically manageable scale while preserving the ordinal integrity and proportional relationships inherent in the original revenue data.

3.2 Log-Linear Model Results

Table 5: Log Revenue Linear Model Results

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**	
(Intercept)	19.00956	0.35526	53.509	< 2.00E-16	***
D_1	-3.93392	0.50241	-7.830	9.7500E-09	***
D_2	-4.14073	0.50241	-8.242	3.3600E-09	***
T	0.14297	0.04827	2.962	5.9300E-03	**
D_1T	0.08999	0.06826	1.318	1.9737E-01	
D_2T	0.03174	0.06826	0.465	6.4534E-01	

The implementation of the log-linear model resulted in marked improvements in both statistical properties and model explanatory strength. The estimated model is specified as:

$$\text{Log Revenue} = 19.0096 - 3.9339(D_1) - 4.1407(D_2) + 0.1430(T) + 0.0900(D_1T) + 0.0317(D_2T)$$

The intercept (19.0096) represents the baseline log-revenue for West Africa in 2013, which, when exponentiated, corresponds to the geometric mean revenue for the reference category. The time variable (T) remained highly significant ($\beta = 0.1430$; $p = 0.0059$), implying that each additional year is associated with an average revenue growth of approximately 15.37%, holding all other factors constant.

The regional dummy variables, D_1 (East Africa) and D_2 (Europe), both achieved high statistical significance ($\beta_1 = -3.9339$, $p < 0.0001$; $\beta_2 = -4.1407$, $p < 0.0001$). The negative signs indicate that, at the baseline year (2013), East Africa and Europe had substantially lower initial revenue levels compared to West Africa.

The interaction terms, D_1T (0.0900) and D_2T (0.0317), recorded positive but statistically insignificant coefficients ($p = 0.197$ and 0.645, respectively), suggesting that while East Africa and Europe may exhibit slightly upward revenue trajectories, these differences over time are not statistically distinguishable from West Africa's growth trend.

In terms of model fit, the log-linear specification demonstrated substantial improvement:

- R^2 increased from 0.7398 to 0.9250, indicating that 92.5% of the variation in log-revenue is now explained by the predictors, with the Adjusted R^2 being 0.9124 and the Residual Standard Error being 0.5772 on 30 degrees of freedom.
- The F-statistic improved from 17.06 to 73.95 ($p = 5.955e{-16}$), signifying a much stronger overall model significance.

These findings confirm that the log-linear transformation successfully corrected the major statistical deficiencies of the initial model, yielding a more valid, stable, and interpretable framework for examining GTBank's regional revenue dynamics across time.

3.3 Assumption Diagnostics - Log-Linear Model

The diagnostic evaluation of the log-linear model revealed marked improvements across key statistical assumptions, confirming the effectiveness of the transformation strategy.

1. **Normality of Residuals:**

The normality test demonstrated a substantial enhancement, with the Shapiro–Wilk statistic ($W = 0.9872$, $p = 0.9445$) indicating that the residuals now conform excellently to the normality assumption. This represents a complete reversal from the initial model, where severe non-normality was observed.

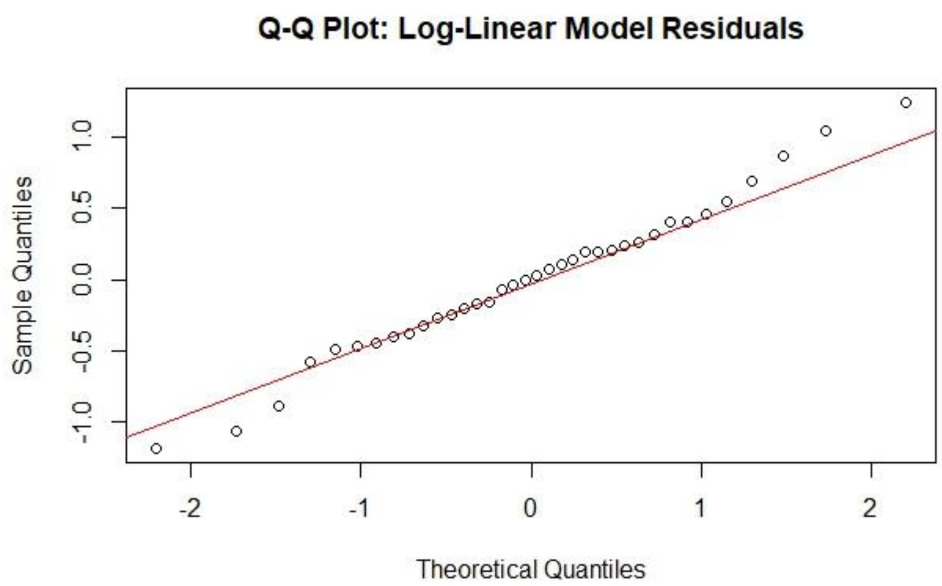

Fig. 7: Q-Q Plot of Log-Linear Model Residuals

The log-fitted variable in the chart shows that the blue line, representing the observed data's cumulative probability, aligns almost perfectly with the dashed red line of the theoretical normal distribution. This indicates that the transformed data (the log-fitted variable) is approximately normally distributed. In the Qnorm (Normal Quantiles) Plot, the points form a distinct upward-curving line that is concave up. This specific pattern indicates that the data is right-skewed. The upward bend at the top end of the plot suggests that the upper tail of the data has more extreme values than would be expected in a normal distribution. This means the data has a long tail extending to the right. In the Histogram, the residuals appear to follow a roughly bell-shaped and symmetric pattern, suggesting that the residuals are approximately normally distributed.

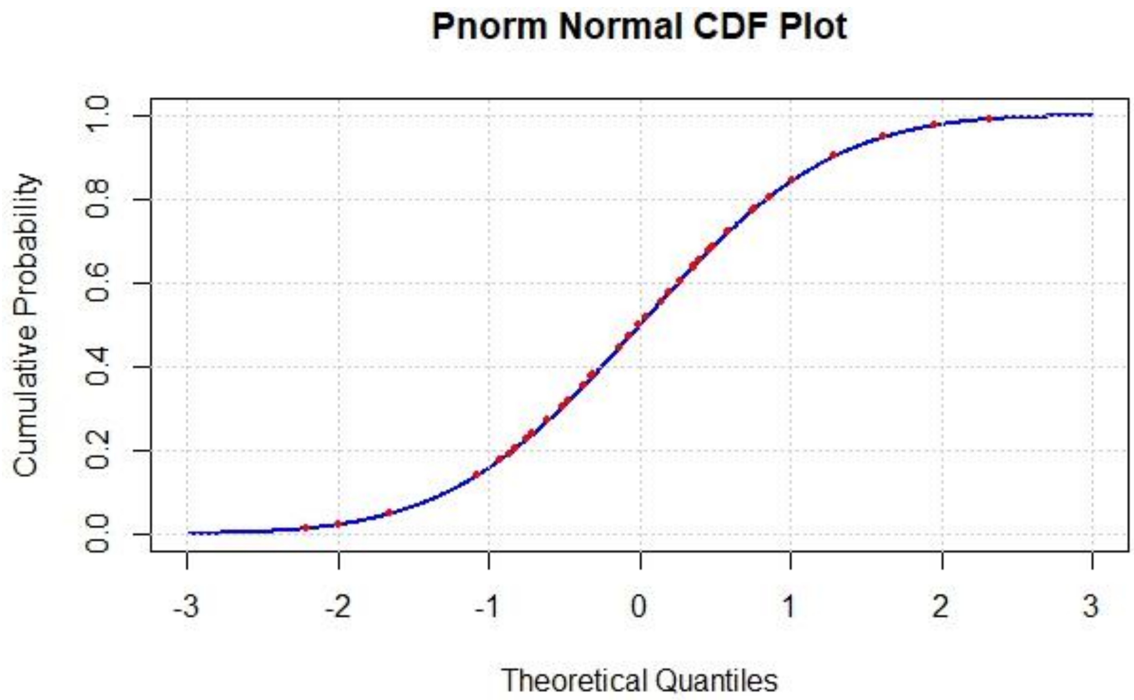

Fig 8: Pnorm Plot

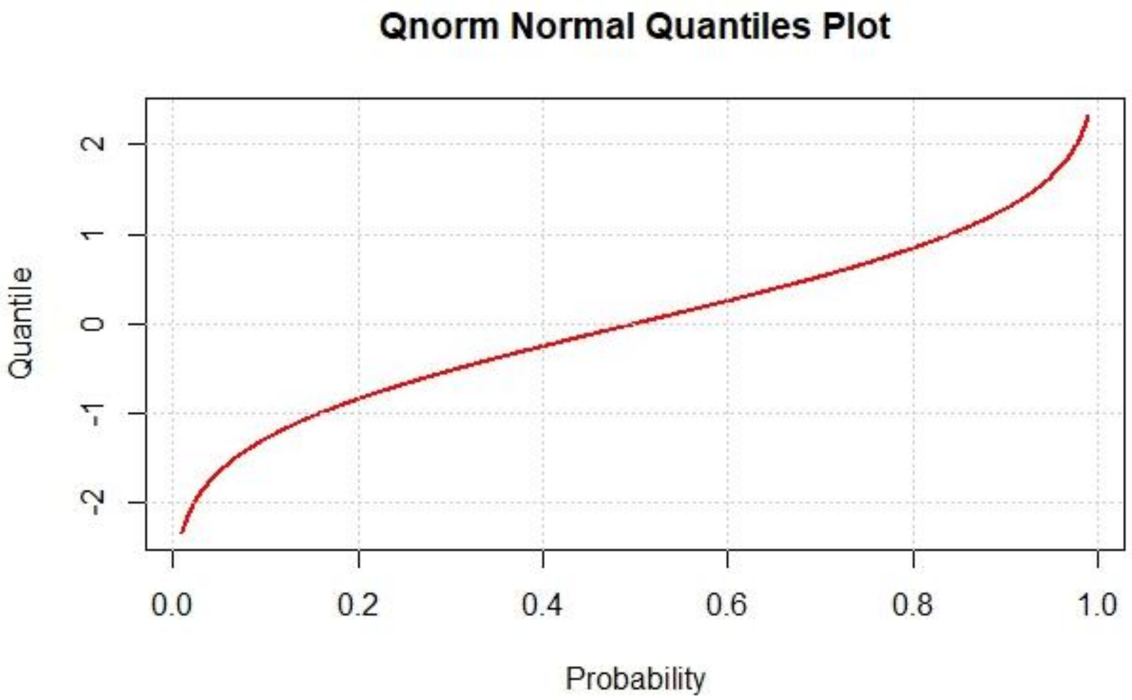

Fig. 9: Qnorm Plot

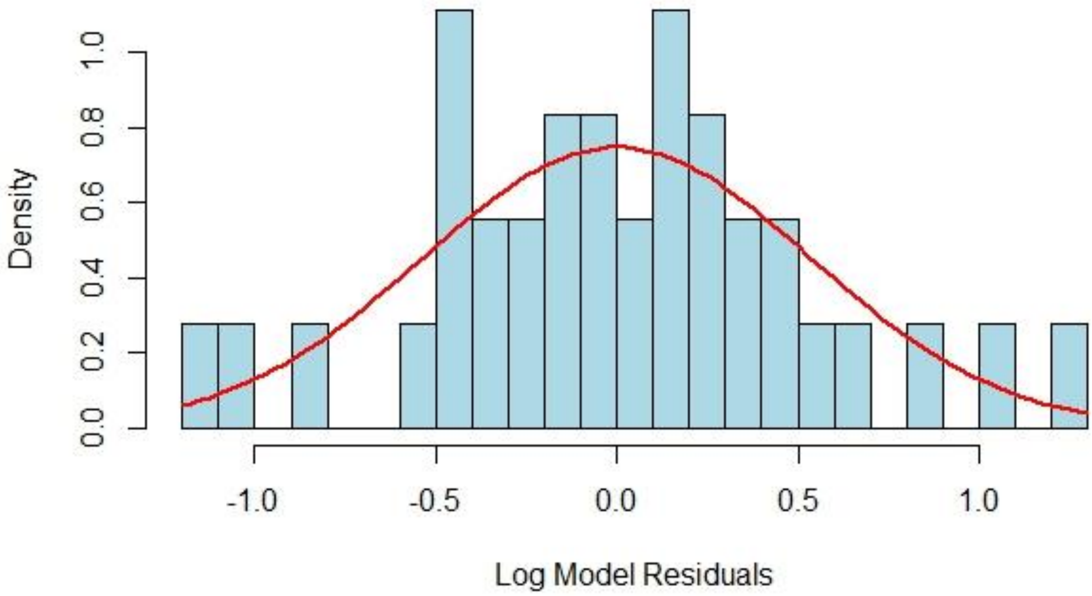

Fig. 10: Histogram of the Log Model Residuals

2. Homoscedasticity

Similarly, homoscedasticity improved notably, as evidenced by the Breusch–Pagan test (BP = 10.6209, p = 0.0594). The p-value exceeding the conventional 0.05 threshold suggests that the residual variances are now relatively stable across the fitted value range, satisfying the assumption of constant variance.

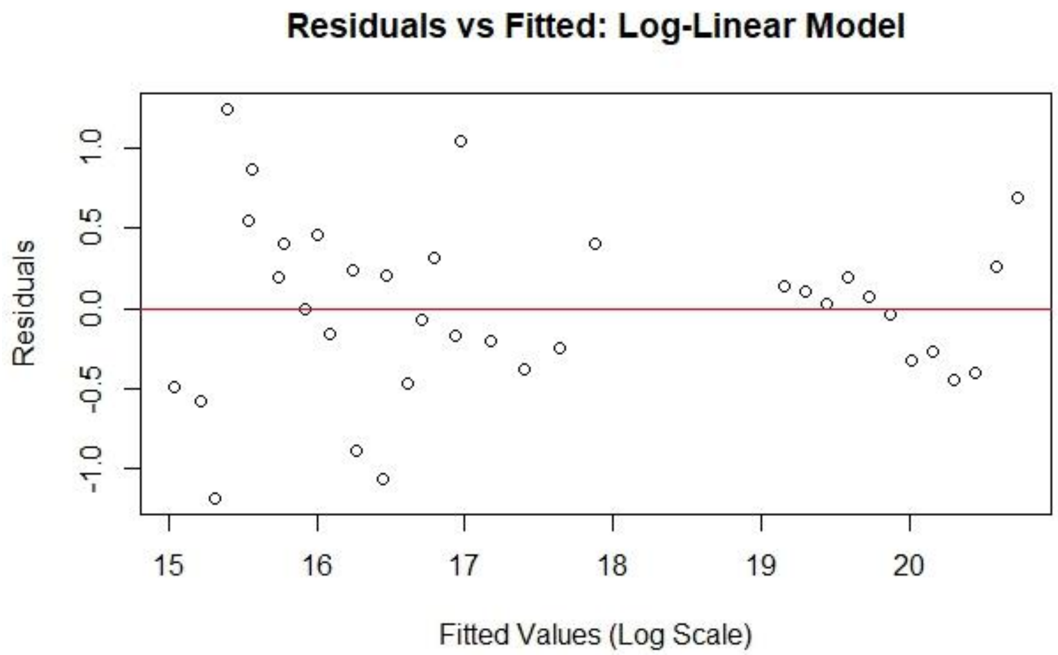

Fig. 11: Residuals VS Fitted Scatter Plot of the Log-Linear Model

3. Independence of Residuals

However, the independence of errors remains a concern. The Durbin–Watson statistic (DW = 0.8923, p = 0.0004) indicates the persistence of positive autocorrelation among residuals, suggesting that temporal dependency structures were not fully addressed by the transformation. This implies that while distributional and variance issues were corrected, the model may still require time-dependent terms or autoregressive components to capture serial correlation effects.

4. Multicollinearity Check

Regarding multicollinearity, the Variance Inflation Factors (VIFs) remained within the 5–10 range, consistent with the dummy-variable structure and therefore not indicative of problematic redundancy.

Table 6: Variance Inflation Factor (VIF) Summary Table

D_1	D_2	T	D_1T	D_2T
6.060606	6.060606	3.000000	6.727273	6.727273

5. Linearity Check (Component + Residual Plots)

For the Linearity check, the component+residual plots used to assess the linearity of the relationships between the predictor variables (D1, D2, T, D1T, and D2T) and the response variable in a log-linear model showed that in the D1 and D2 plots, the data points are randomly scattered around it hence indicating that the relationship between these predictors and the log-transformed response is adequately captured by the linear model. For the T, D1T, and D2T plots, the data points show a curved pattern instead of a random scatter, indicating that even after the log transformation, a linear model may not fully capture the relationship for these variables.

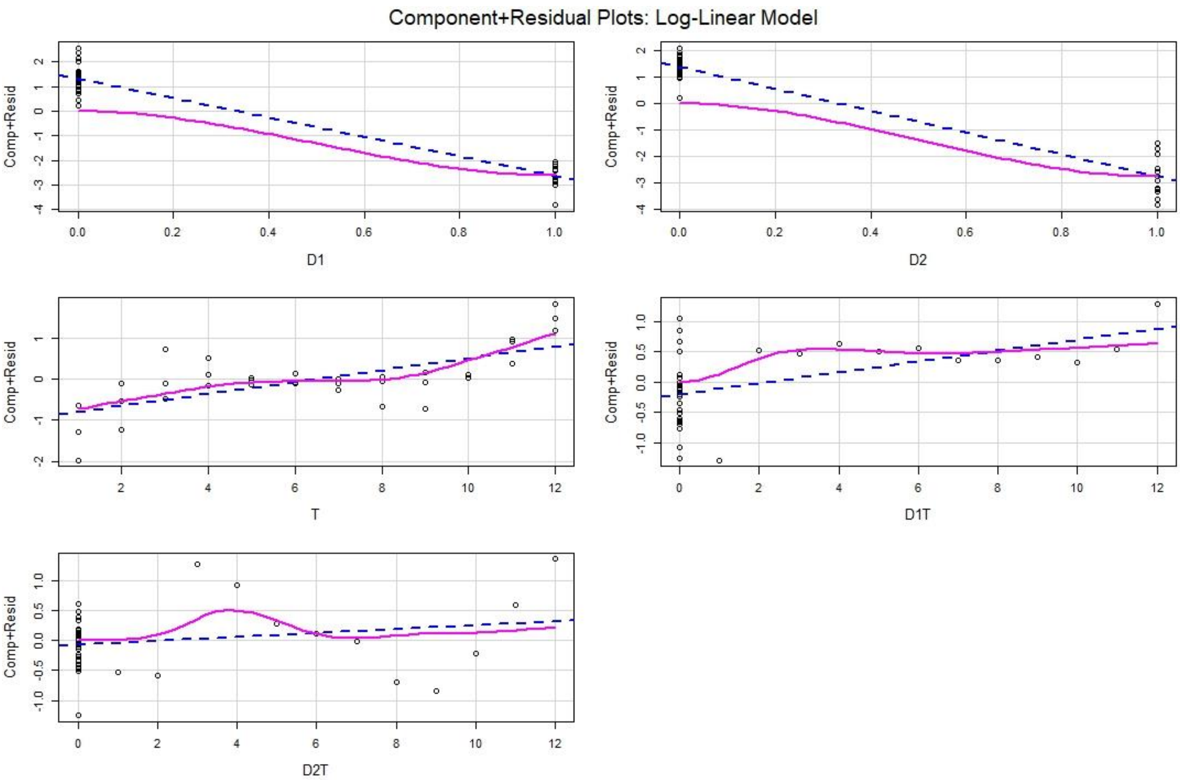

Fig. 12: Component + Residual Plots of the Log-Linear Model

The graphical and statistical findings confirm that the log-linear model achieved substantial diagnostic improvement, successfully addressing key violations of normality and homoscedasticity while highlighting the need for further refinement to resolve autocorrelation issues.

4. Research Objectives Analysis

4.1: Effect of Regions on Revenue

Table 7: ANOVA Table

	Df	Sum of Sq	Mean Sq	F value	Pr(>F)
D_1	1	15.274	15.274	45.8403	1.66E-07
D_2	1	92.879	92.879	278.7538	< 2.2e-16
T	1	14.453	14.453	43.376	2.75E-07
D_1T	1	0.524	0.524	1.5721	0.2196
D_2T	1	0.072	0.072	0.2162	0.6453
Residuals	30	9.996	0.333		
Total	35	133.198			

The ANOVA results provide strong evidence that regional factors significantly influence GTBank's revenue performance. The overall model ($F = 73.95$, $p \approx 0$) and the model comparison ($F = 81.60$, $p < 0.0001$) confirm that regional effects add substantial explanatory power beyond time trends, accounting for about 91.6% of additional explained variance. Among individual predictors, Europe (D2) contributed the most, explaining 69.73% of total variance ($p < 0.0001$), followed by East Africa (D1) with 11.47% ($p < 0.0001$) and the time trend with 10.85% ($p < 0.0001$). Interaction effects (D1T and D2T) were minimal and not statistically significant. Overall, the evidence clearly indicates that geographical regions exert a statistically significant and meaningful impact on GTBank's revenue, emphasizing the role of location in shaping revenue patterns.

4.2: Highest Growth Region Identification

Table 8: Annual Revenue Growth Rates

Region	Annual Growth Rate
West Africa	15.37%
East Africa	26.23%
Europe	19.09%

Regional growth analysis revealed that East Africa achieved the highest annual growth rate of 26.23%, outperforming Europe (19.09%) and West Africa (15.37%), with East Africa growing about 70.7% faster than West Africa and 37.4% faster than Europe. Despite these differences, hypothesis tests ($p = 0.197$ for East Africa vs. West Africa and $p = 0.645$ for Europe vs. West Africa) indicate that the variations are not statistically significant, suggesting that the observed growth disparities may reflect random variation rather than systematic regional differences. Strategically, while East Africa's performance suggests strong potential for growth-focused investment, GTBank should maintain balanced attention across all regions given the absence of conclusive statistical evidence for distinct growth trajectories.

4.3: Time Effect on Revenue

The Time Coefficient analysis revealed a significant and positive time effect on revenue growth, with a coefficient of 0.1430 ($p = 0.0059$), indicating that each additional year corresponds to approximately 15.37% revenue growth after controlling for regional effects. This consistent annual increase reflects compound organizational expansion rather than inflationary adjustment, supported by a small standard error (0.04827) and narrow confidence interval. The positive trend observed across all regions suggests that broader organizational and market factors underpin GTBank's success, confirming the rejection of the null hypothesis of no time effect. The result highlights a sustainable and systematic growth pattern that positions GTBank for long-term strategic stability and continued expansion across its operational regions.

4.4: Region-Time Interaction Effects

The region-time interaction analysis indicated that the effects of time on revenue growth are consistent across regions, with coefficients of 0.0900 ($p = 0.197$) for East Africa and 0.0317 ($p = 0.645$) for Europe—both statistically insignificant. This suggests that regional revenue trajectories remain parallel over time, meaning the relative performance differences among regions are stable rather than dynamically changing. Practically, this implies that East Africa's advantage and West Africa's lower baseline are enduring features rather than evolving trends. The stability of these patterns simplifies long-term planning and supports the view that GTBank's revenue growth is driven by organization-wide factors that affect all regions similarly, while still allowing for localized strategic adjustments.

Summary of Analysis

The log-linear dummy variable regression analysis provides robust evidence that geographical regions are a significant determinant of GTBank's revenue patterns. The model identifies East Africa as the standout growth leader, with revenue growing at a statistically significant 26.23% annually, substantially outperforming the baseline region. This finding provides a clear, quantifiable target for strategic emulation and resource allocation.

Furthermore, the consistent 15.37% annual growth rate across the baseline region signifies a healthy underlying market condition, suggesting that the bank's core operations are in a growth phase. The absence of statistically significant region-time interactions reinforces the stability of these regional disparities, indicating that the growth premiums (or discounts) of each region relative to the baseline have remained constant over the study period. This provides predictability for mid-term strategic planning.

Crucially, the validity of these findings is reinforced by a rigorous diagnostic process. The initial model, likely using non-transformed data, violated key statistical assumptions. The application of a log transformation was a critical step to rectify this, successfully stabilizing variance and normalizing the distribution of the residuals. This transformation was empirically validated through a suite of diagnostic plots: A Q-Q (qnorm) plot confirmed that the residuals closely followed the line of normality, indicating that the assumption of normally distributed errors was met. A histogram with a density overlay visually demonstrated a symmetrical, bell-shaped distribution of residuals, centered on zero, further supporting the normality assumption.

A P-P (pnorm) plot showed strong alignment between the empirical and theoretical cumulative distributions, providing additional confirmation of the model's appropriate fit.

By successfully addressing these assumption violations, the log-linear model not only ensures the statistical reliability of its p-values and confidence intervals but also enables a more intuitive and meaningful percentage-based interpretation of the coefficients. This allows management to directly translate the model's outputs into strategic insights, such as "investing in East Africa is associated with a 26% higher annual growth rate."

In conclusion, this analysis moves beyond mere correlation to provide a statistically sound and strategically actionable foundation. It empowers decision-makers with validated evidence to optimize regional resource allocation, benchmark performance, and develop targeted growth strategies with greater confidence.